

Process intensification of catalytic-wall Taylor-Couette reactors by reinforcement learning

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1 Abstract

Previous work intensified reactions in catalytic-wall Taylor-Couette reactors by modulating the angular velocity of the reactor’s inner cylinder, saving a considerable amount of energy [4]. Here we extend their methods by creating a RL system that chooses unconventional modulation strategies that requires less energy than previous benchmark policies.

2 Introduction

Process intensification tends to be an expensive endeavor implying replacing old equipment with new and improved ones. Here we offer a treat giving you similar scale of improvement without asking for your company credit score. Taylor-Couette reactors are super predictable and consistent so they have become very popular in industrial processes such as crystallization, electrolysis, and catalysis.

3 Methodology

3.1 Problem Formulation

For the sake of knowing whether our RL approach outperforms Lopez-Guajardo et al. [4], our formulation closely follows theirs. The Taylor-Couette reactor geometry is summarized in Table 1.

Table 1: Geometry of the catalytic-wall Taylor-Couette reactor. The computational domain is a 5° azimuthal wedge of the full annulus, with a one-cell side-outlet band at the bottom of the outer wall.

Parameter	Symbol	Value
Inner cylinder radius	R_i	25.40 mm
Outer cylinder radius	R_o	31.75 mm
Gap width	$d = R_o - R_i$	6.35 mm
Reactor height	H	38.10 mm
Radius ratio	$\eta = R_i/R_o$	0.80
Aspect ratio	$\Gamma = H/d$	6.0
Side-outlet band height	h_{out}	0.42 mm
Azimuthal wedge angle	θ	5°

The incompressible Navier-Stokes equations

$$\rho \nabla \cdot \mathbf{u} = 0, \quad (1)$$

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \mu \nabla^2 \mathbf{u}, \quad (2)$$

and the transport of dilute species equation

$$\frac{\partial c}{\partial t} + \mathbf{u} \cdot \nabla c = D \nabla^2 c, \quad (3)$$

were solved by FVM with the OpenFOAM v2506 [6] framework.

The Reynolds number was defined as $Re = \frac{\omega R_i d}{\nu}$, where ω is the angular velocity of the inner cylinder and ν is the kinematic viscosity of the fluid. The flow was assumed to be laminar and incompressible, and the no-slip boundary condition was applied at both cylinder walls.

The Farama Foundation’s Gymnasium [5] was used as the framework for our reinforcement learning (RL) environment with

4 Results

5 Discussion

References

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